A robot that acts to know

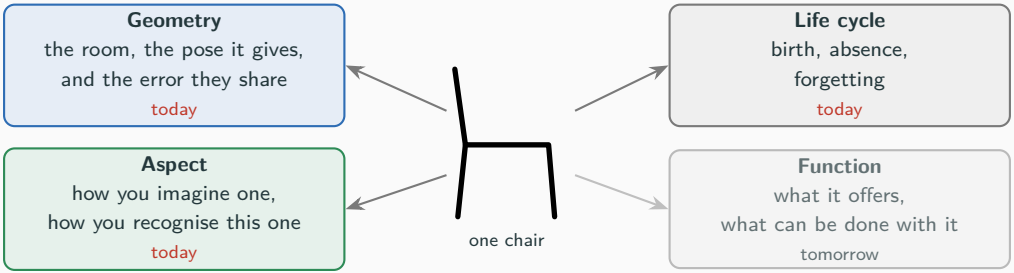
Day 3 — Concepts: geometry, life cycle, aspect

Course built on the doctoral thesis of Noé Zapata

Wednesday 2 December

Montevideo, 30 November – 4 December 2026

Yesterday: a concept has dimensions. Today: three of them

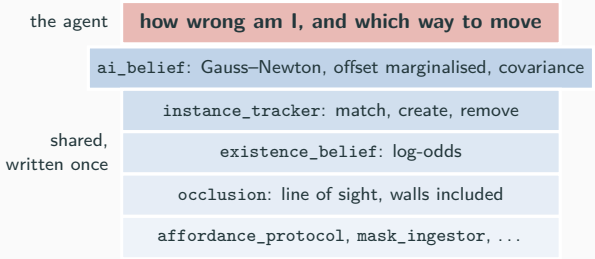


First, the thing that answers for all of them: the concept agent.

source: the dimensions are this course's way of laying out the thesis

What a concept agent is

An agent writes two things

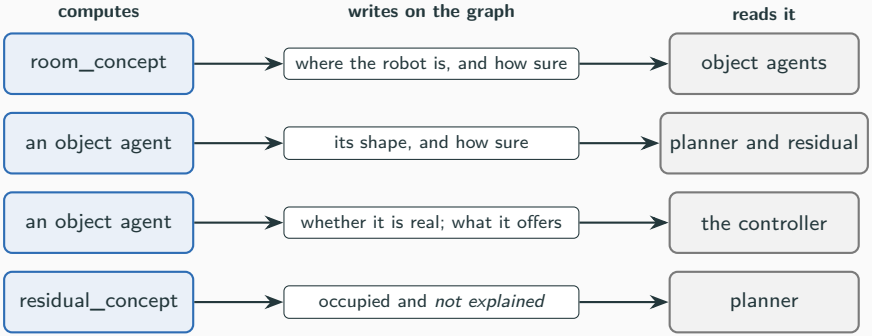


A residual and its Jacobian. All the Bayesian mathematics lives in the engine, once.

This is why a roster entry can be half a page: bottle needs little, table and refrigerator fill the template, and that is a fact about the objects.

source: concept_agents.tex l.534–537, l.136–138, l.1196–1203

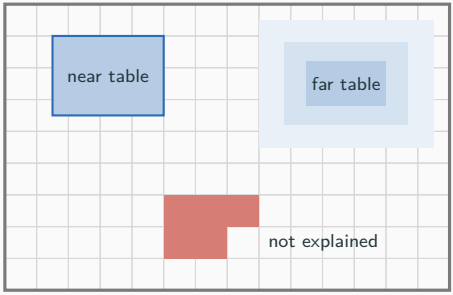
Each publishes what only it can compute



Nobody asks anybody. Everything an agent knows about another's work is on the graph.

source: concept_agents.tex l.1209–1216, l.1533–1544

residual_concept: whatever nobody else explains



a sure table claims its cells tightly; an unsure one loosely

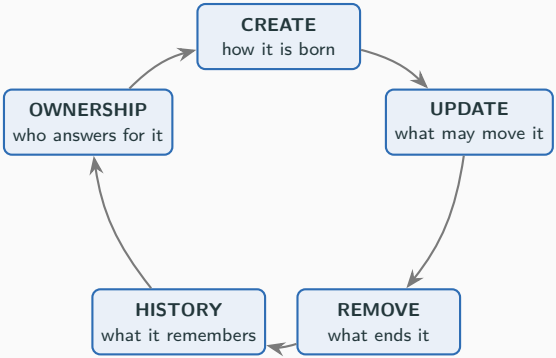
A grid of what the sensors hit, minus what every other agent claims to explain. What is left is what the robot must avoid.

Nothing has to be recognised for the robot not to hit it.

No safety margin anywhere: each claim is as wide as the claimant's own published uncertainty. That is why the others must publish it.

source: concept_agents.tex l.1497–1538

Every agent answers to a common contract



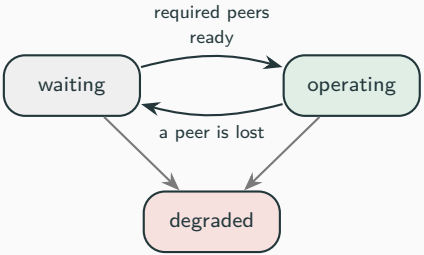
Not paperwork: this is what anything that integrates noisy fragments has to do.

A rule written in a contract is a rule a new agent cannot forget.

Next: UPDATE is geometry; CREATE and REMOVE are the life cycle.

source: concept_agents.tex l.169–180; CONCEPT_AGENT_LIFECYCLE.md l.7–13

The factorisation is only true while its owners are alive



Specified in TLA$^+$ and model-checked: 8 safety and 4 liveness properties, no counterexample.

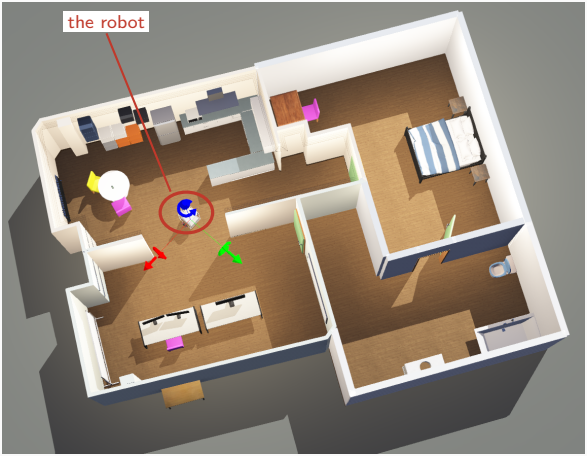
A process crashes and its node stays. It looks like any other data, and nobody is keeping it true any more.

A peer that is present is *running*, which is not the same as *able to serve*. Heartbeats are ordinary graph nodes.

source: cortex.tex l.732–741, l.757–775, l.780–785

Geometry

The world: three rooms the robot has never seen



the simulated apartment (Webots): three rooms, furniture, doors

At the start the robot knows nothing about this place: no walls, no furniture, no doors.

CORTEX gives it an intrinsic drive: build a model of the environment, one room at a time, as close to the real one as it can.

"As close as it can" has a measure, the free energy: how badly the model explains the evidence, plus the price of the model's own detail.

Today: the room, the robot's place in it, one object inside it (a chair), and a glance at the doors between rooms.

The room as a probability model (1/4): what is unknown, what is given

$o_{1:t}$ the laser beams: cells crossed, where each stopped

$X_{0:t}$ the robot's poses, from the start

$\delta_{1:t}$ what the wheels measured, step by step

m the room's *structure*: corners, and the turns between them

θ_m the corners' coordinates, for that structure

$$p(o_{1:t}, X_{0:t}, \theta_m, m, \delta_{1:t})$$

condition on δ

$$p(o_{1:t}, X_{0:t}, \theta_m, m \mid \delta_{1:t})$$

the joint: one number for every combination of all five

why: δ is the robot's own wheel reading. It is always there, and nothing in the room explains it. Conditioning means: take it as given, and spend no factor on how likely it was.

unknown: X, θ_m, m
to be inferred; for m, even how many corners (frame 12)

observed, left of the bar: o
measured, and the model must say how likely it is

given, right of the bar: δ
measured, and never explained

The room as a probability model (2/4): factorise, exactly

Day 1's rule, with something known. A known E stays behind the bar in every factor:

$$P(A, B \mid E) = P(A \mid B, E)\, P(B \mid E)$$

and, applied three times,

$$P(A, B, C, D \mid E) = P(A \mid B, C, D, E)\, P(B \mid C, D, E)\, P(C \mid D, E)\, P(D \mid E)$$

exact, always, and in any order of A, B, C, D

The room: $A, B, C, D = o, X, \theta_m, m$ and $E = \delta$ (time indices dropped).

$$p(o, X, \theta_m, m \mid \delta) = \underbrace{p(o \mid X, \theta_m, m, \delta)}_{\text{the scans}}\ \underbrace{p(X \mid \theta_m, m, \delta)}_{\text{the path}}\ \underbrace{p(\theta_m \mid m, \delta)}_{\text{the corners}}\ \underbrace{p(m \mid \delta)}_{\text{the shape}}$$

We pick this order because, read right to left, it is how the data come about: a shape, its corners, a path through the room, and the scans taken along it.

Still exact: nothing has been assumed yet.

The room as a probability model (3/4): assume, by crossing out

$p(o \mid X, \theta_m, m, \cancel{\delta})$	poses and room known: the wheels add nothing about the scans
$p(X \mid \cancel{\theta_m, m}, \delta)$	the path follows the wheels, not the room (only roughly: walls limit it)
$= \prod_\tau p(X_\tau \mid X_{\tau-1}, \delta_\tau)$	and each pose needs only the previous one and that step's wheels (τ: one step)
$p(\theta_m \mid m, \cancel{\delta})$	the wheels say nothing about the corners
$p(m \mid \cancel{\delta})$	... nor about the room's shape

$$\Downarrow$$

$$p(o, X, \theta_m, m \mid \delta) = \underbrace{p(o \mid X, \theta_m, m)}_{\substack{\text{the scans, given the} \\ \text{poses and the room}}} \underbrace{\prod_\tau p(X_\tau \mid X_{\tau-1}, \delta_\tau)}_{\substack{\text{the motion,} \\ \text{given the wheels}}} \underbrace{p(\theta_m \mid m)}_{\substack{\text{the corners,} \\ \text{given the shape}}} \underbrace{p(m)}_{\text{the shape}}$$

The chain rule is always exact. The model is what we choose to cross out.

source: layout_acquisition.tex l.101–124 (eq. la-joint); AuRo paper eq. joint, the "robot and room" block

The room as a probability model (4/4): Bayes turns the question around

the model (frames 8–10): how a room, a path and their scans come about together



a room (m, θ_m) and a path (X)

inference: these scans were read; which rooms and paths explain them?



the scans (o), as read

$$\underbrace{p(X, \theta_m, m \mid o, \delta)}_{\text{what we want: } o \text{ now behind the bar}} = \frac{p(o, X, \theta_m, m \mid \delta)}{\underbrace{p(o \mid \delta)}_{\text{one number once } o \text{ is read, the same for every room}}} \qquad \textbf{exact Bayes}$$

$$\propto \underbrace{p(o \mid X, \theta_m, m)}_{\textbf{likelihood}\text{: how likely these scans, if the room and path were these}} \cdot \underbrace{\textstyle\prod_\tau p(X_\tau \mid X_{\tau-1}, \delta_\tau)\, p(\theta_m \mid m)\, p(m)}_{\textbf{prior}\text{: how likely that path and room, before any scan}} \qquad \textbf{model approximation}$$

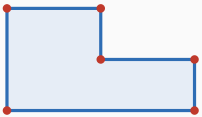
Posterior $\propto$ likelihood $\times$ prior: to compare two rooms, evaluate it for each.

$\propto$: equal up to a constant factor. Its $-\ln$ is step 5's free energy.

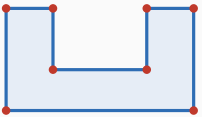
What could the room be?



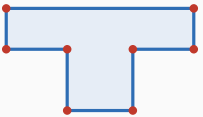
a rectangle: 4 corners



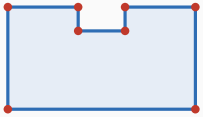
an L: 6 corners



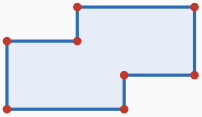
a U: 8 corners



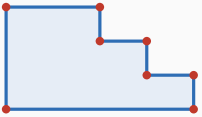
a T: 8 corners



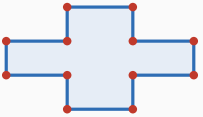
an alcove: 8 corners



a step: 8 corners



a staircase: 8 corners



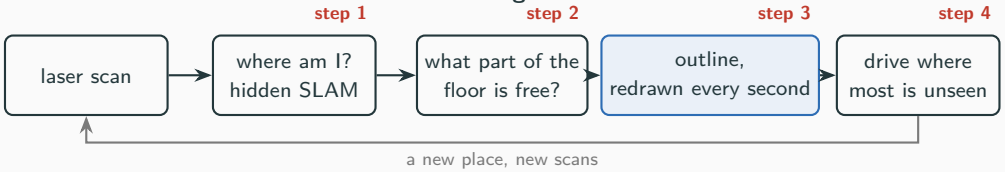
a cross: 12 corners

Every one is a right-angled (*Manhattan*) room, and the robot does not know which. From one corner an L looks like a rectangle, and a U like an L.

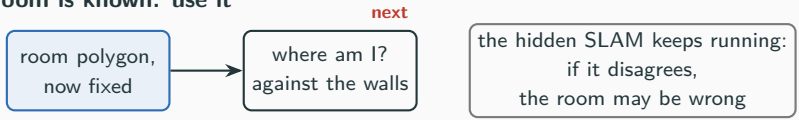
The number of corners is unknown, so even the size of the answer is unknown: the robot must choose the shape, not just fit its numbers. Step 5 prices each corner.

Estimating a room: the whole procedure

1. The room is unknown: build it while driving



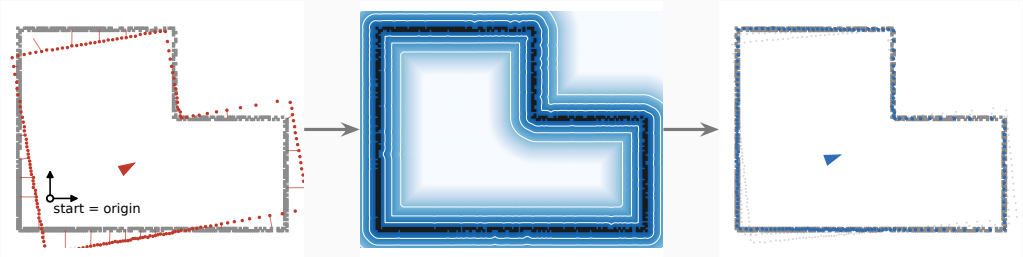
2. The room is known: use it



While the room is unknown, the outline never corrects the pose: a wrong wall would pull the pose, and the pulled pose would confirm the wrong wall.

source: layout_acquisition.tex l.68–82, l.162–199

Step 1: a hidden SLAM. Where am I, before any wall exists?



1. **Predict** where the wheels say: 30 cm, 10° off

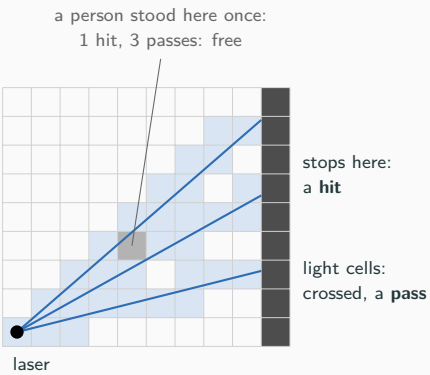
2. **The map**: the grid's hits, as distances; darker = closer to a hit

3. **Correct** until the returns sit on hits: under 1 cm off; then add the scan

Correct = find the pose X that minimises $\sum_n D\big(\text{return } n \text{ placed at } X\big)^2$, with D the middle map.
Gauss–Newton: the slope of D under each return says which way to move X.

source: layout_acquisition.tex l.344–416; computed in `figures/make_d3_slam.py`

Step 2: the same grid also counts passes. What part of the floor is free?



Hits say where things *are*. They cannot say where *nothing* is, and the room is exactly that: the free floor the robot can reach.

So each 5 cm cell keeps two counts, every beam followed cell by cell:

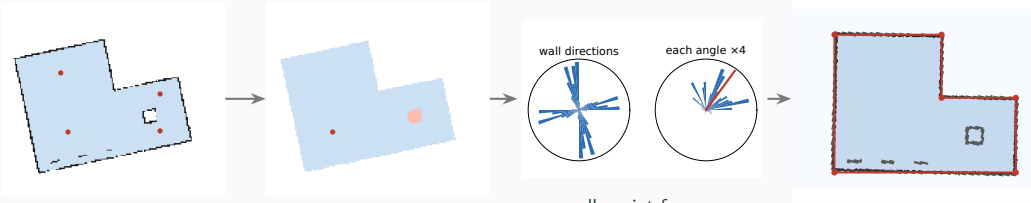
- **passes**: beams that crossed it (empty);
- **hits**: beams that stopped in it (something is there).

More hits than passes: **matter**. Otherwise, if crossed: **free**.

Step 1 reads only the hits; the outline reads both.

source: layout_acquisition.tex l.507–524

Step 3: from the grid to an outline, every second



a. the grid after four scans (red): light = free, black = hit

b. the free region around the robot; the wardrobe's hole is filled (pink)

c. walls point four ways, 90° apart; each angle ×4, then mod 360°: they land together; average, ÷4: the axis, 13° (true: 12°)

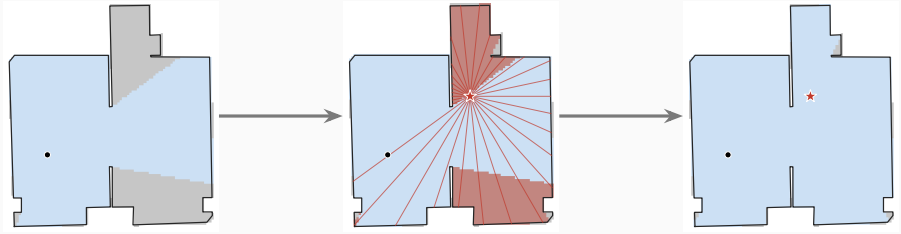
d. in that axis, the region's border (black) straightened into horizontal and vertical edges: 6 corners. Cabinets and wardrobe (grey) stay *inside*

A simulated L room, turned 12°, with cabinet fronts along one wall and a wardrobe.

Redrawn from scratch every second, with no memory: a proposal, not yet the room (step 5 judges it).

source: layout_acquisition.tex l.526–560; free_space_polygon.h l.433–620; `figures/make_d3_outline.py`

Step 4: drive where most is unseen; stop when nothing is



1. from the start (•): floor seen (blue), hidden (grey)

2. go where most hidden floor would show (⋆, red)

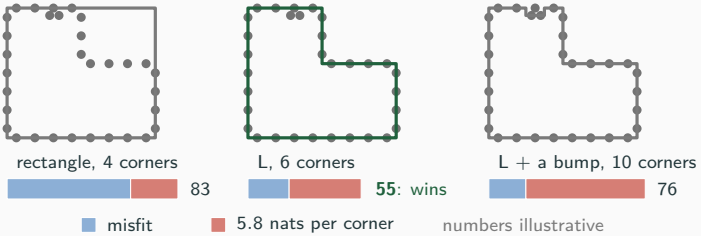
3. what is left is smaller than the robot: **stop**

The non-convexity problem: the walls that jut into the room hide floor, so no single place sees it all. To know the room, the robot has to move.

The epistemic term of Day 4 in its simplest form: unseen floor stands in for information.

source: room R1, layouts/apartamento_layout.svg; layout_acquisition.tex l.663–720; figures/make_d3_explore.py

Step 5: which outline becomes the room? Bayes, on the structure



$$p(m \mid o) \;\propto\; p(o \mid m)\, p(m), \qquad p(o \mid m) = \int p(o \mid \theta_m, m)\, p(\theta_m \mid m)\, d\theta_m$$

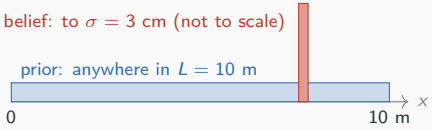
$$-\ln p(o \mid m) \;\approx\; \underbrace{-\ln p(o \mid \hat{\theta}_m, m)}_{\text{misfit of the best-placed outline } (\hat{\theta}_m)} \;+\; \underbrace{5.8\,\text{nats} \times \text{corners}}_{\text{the price of averaging over where the corners could be}} \;=\; F(m)$$

Bayes as in frame 11, now over the *structure* m: lowest F = most probable structure.
nat: the unit of $-\ln p$. A difference of k nats is odds of e^k to 1 (5.8 nats $\approx$ 333 to 1).

source: layout_acquisition.tex l.1033–1075 (eq. la-mdl), l.56–66

Step 5, continued: the price of a corner

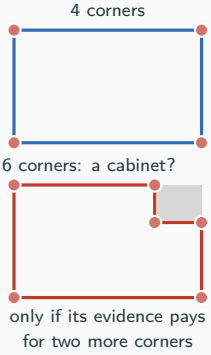
Why does a corner cost 5.8 nats?



The price of one coordinate is the complexity term (Day 1), the KL divergence between the two uniforms $\mathcal{U}$:

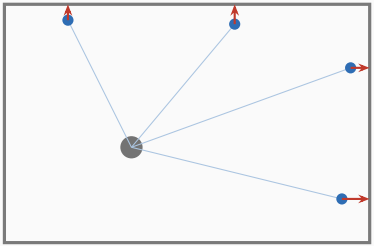
$$D_{\mathrm{KL}}[\mathcal{U}_\sigma \,\|\, \mathcal{U}_L] = \int_\sigma \tfrac{1}{\sigma} \ln \frac{1/\sigma}{1/L}\, dx = \ln \frac{L}{\sigma} = \ln \frac{10}{0.03} \approx 5.8 \,\text{nats}$$

Choosing one of 333 slots. In a right-angled room each new corner adds one new coordinate, so a corner costs 5.8 nats, and it is kept only if the evidence pays for it: the trade an object will face at birth.



source: layout_acquisition.tex l.56–66, l.128–146

The room supplies the likelihood



each return is pulled onto its nearest wall

Where would I have to stand for this scan to land on these walls?

- The distance from each return to its wall is the prediction error. No expected scan is ever drawn.
- More points in a scan do not make the belief more certain: their errors are not independent.
- Range and incidence weigh each point: precisions, not gates.

source: room_concept.tex l.243–307

Localising: Kalman's structure, but optimised

Day 1: with a Gaussian prior and a linear sensor, minimising F *is* the Kalman update. Localising has the same two parts:

$$F(X) \;=\; \underbrace{\tfrac{1}{2}\,(X-\bar{X})^{\top}\bar{P}^{-1}(X-\bar{X})}_{\text{predict: where the wheels say}} \;+\; \underbrace{\tfrac{1}{2\sigma^2}\textstyle\sum_n d_n(X)^2}_{\text{update: every return onto its wall}}$$

X	the pose: x, y, heading	$d_n(X)$	distance from return n, placed at X, to its wall
$\bar{X}$, $\bar{P}$	pose predicted by odometry, its covariance	σ	noise of one such distance

Kalman (extended): linearise d_n once, at $\bar{X}$, and take one step.

That single Gauss–Newton step from $\bar{X}$ *is* the EKF update: the same gain, the same covariance.

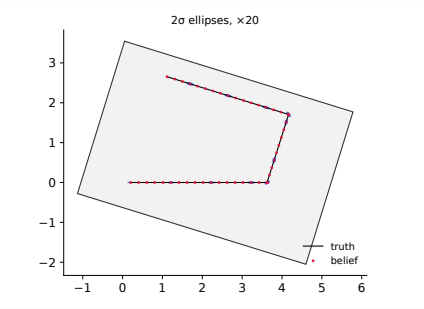
Here: keep stepping, re-linearising at each new pose, until F stops falling. It matters when the heading is off: a wall distance turns with it.

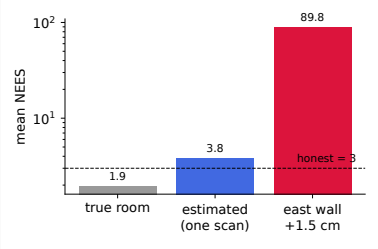
Covariance $P = \mathcal{H}^{-1}$, with $\mathcal{H} = \bar{P}^{-1} + J^{\top}J/\sigma^2$ the curvature of F at its minimum and J how each d_n changes with the pose: Day 1's Laplace.

Prior from motion, likelihood from the sensor: optimised, not linearised once.

source: room_concept.tex l.243–300, l.1601–1640; Day 1, the Kalman update as free energy

Localising against the room you just built





A wall **1.5 cm** out of place makes the localiser confidently wrong.

NEES: squared error ÷ claimed variance, summed over x, y, heading. Honest: 3. Here: 90.

In the operating regime the polygon is a constant, so its error is shared by every scan, for ever.

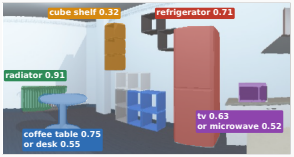
source: lab 3, exercise 13

What an object agent sees: a mask



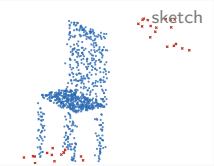
1. The camera
colour and depth, every frame

segment



2. Retina's segmenter
one mask per object,
with a label and a score

depth
+ pose $\hat{X}$



3. A chair's mask in 3-D
dots: the chair;
crosses: spill onto floor, wall

label + score	**points**	**variance**	**bias**
a guess: table 0.75, or desk 0.55	each with its own σ_n, growing with range	blur and jitter, zero mean: it averages out	the camera turned: all points shift one way, and it does not average out

A mask says "chair", not *which* chair: each one is handed to an instance, or starts a new one. What instance i was handed is its data $\bar{Z}_i$, and the handing is one best guess (frame 26).

source: retina, `yoloe-26x-seg-household`, on a simulated frame; phylogenetic_perception.tex l.347–408

The chair: the same free energy

Now the robot's pose is known, from the localiser, and the chair's is not. Bayes as for the room (frame 11): posterior $\propto$ prior $\times$ likelihood, and F is its $-\ln$:

$$F(\theta) \;=\; \underbrace{\tfrac{1}{2}\,(\theta-\bar{\theta})^{\top}\,\bar{P}^{-1}(\theta-\bar{\theta})}_{\text{prior: where I believed the chair was}} \;+\; \underbrace{\textstyle\sum_n \rho\big(d_n(\theta)/\sigma_n\big)}_{\text{likelihood: every mask point onto the chair}}$$

θ	the chair's pose: x, y, heading	$d_n(\theta)$	mask point n's distance to the chair
$\bar{\theta}$, $\bar{P}$	last belief, covariance; widened when unseen	σ_n	point n's noise; grows with range

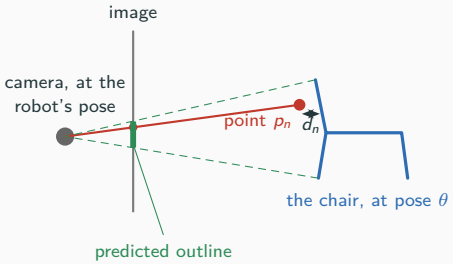
$\rho(u) = -\ln p(u)$: a point's cost at u standard deviations. A Gaussian gives $u^2/2$; "chair or clutter" stops growing, so a stray point pulls with bounded force.

The shape is not fitted: it is the class template, seat 60 $\times$ 52 cm with its top at 0.60 m and a backrest 0.66 m above it. Only the pose is estimated.

Minimised by the same Gauss–Newton steps as the robot's pose; the covariance is again the inverse curvature, $P = \mathcal{H}^{-1}$ (Day 1's Laplace).

source: concept_agents.tex l.469–535; chair_concept: chair_dof.h (pose-only), common/concept_manifest/chair.concept.toml

The chair's likelihood: signed distance, mixture, reprojection



Signed distance. Each mask pixel, with its depth, is placed in the room through the robot's pose: a 3-D point p_n. Its error d_n is its distance to the chair's surface: 0 on it, + outside, − inside.

Mixture. The chair is a seat, a backrest and four legs. A point is either on the chair or clutter, $\rho = -\ln(\text{chair} + \text{clutter})$, so a stray point costs a bounded amount.

Precision. σ_n grows with range. One error is shared by all the points of a frame, the robot's own pose error, and it is counted once (frame 27).

Reprojection. The reverse path: the believed silhouette, projected into the image with its predicted depth, is checked against the edges there. On the robot it feeds existence, not the fit.

source: concept_agents.tex l.516–523; chair_concept/src/chair_fitter.cpp, chair_contour.h

Association: the mistake the free energy cannot see



a poorly known object accepts detections from further away

Each detection goes to the object it most likely belongs to: one best guess, closest first.

A guess has no uncertainty to pay for, so it adds nothing to F. A wrong match is invisible to the quantity that would have registered it.

The repair comes later and from elsewhere: merging, and the existence belief.

source: concept_agents.tex l.483–486, l.859–889

The common-mode ceiling



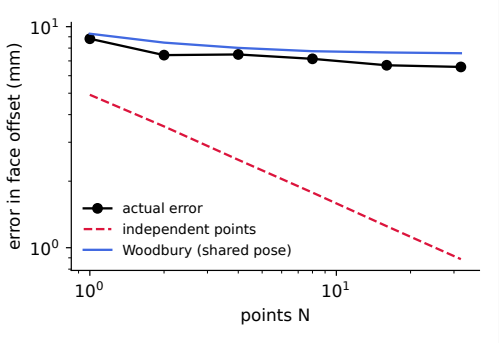
The same arrow as yesterday, now on a thousand points.

Averaging points removes their small independent jitter. It cannot touch the arrow.

A denser mask eventually stops helping: the limit is set by the robot's pose, not by the number of pixels. And the data cannot tell a displaced object from a displaced robot.

source: concept_agents.tex l.1048–1090

The ceiling, measured in the toy



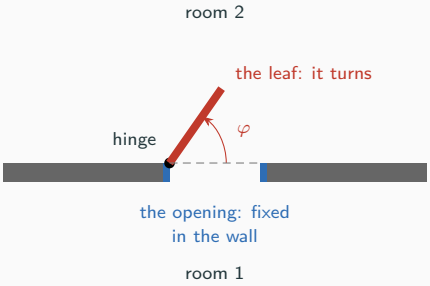
The robot maps the table's face from a *dead-reckoned* pose.

- Treating points as independent claims under 1 mm at $N = 32$.
- The real error stalls near 7 mm.
- Woodbury predicts it to within about 15% at every N.

Averaging removes independent error. It does not remove shared error.

source: lab 3, exercise 15

Doors: the same idea, with a moving part



Estimated like the chair, a prior and a likelihood minimised, but articulated: two parts, and one of them moves.

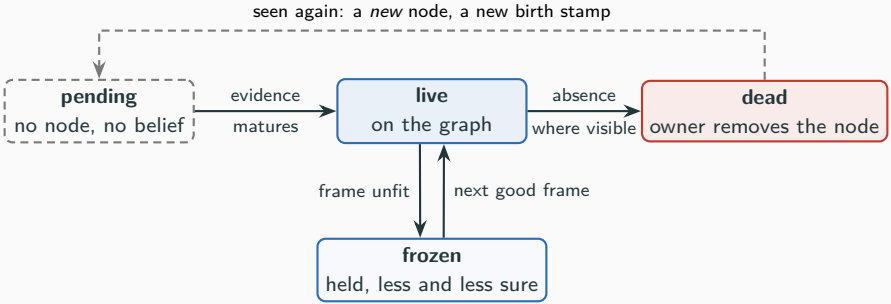
- **The opening**: where along its wall, how wide, how high; from the door masks.
- **The leaf**: its angle φ, from the lower laser. Someone may move it while the robot is away, so a remembered angle loses weight with time.
- Which way it swings decides where the robot may stand to open it.

A door also joins two rooms: crossing it changes which room the robot is in. We stop here.

source: crossing.tex l.307–338 (the door as an estimate)

Life cycle

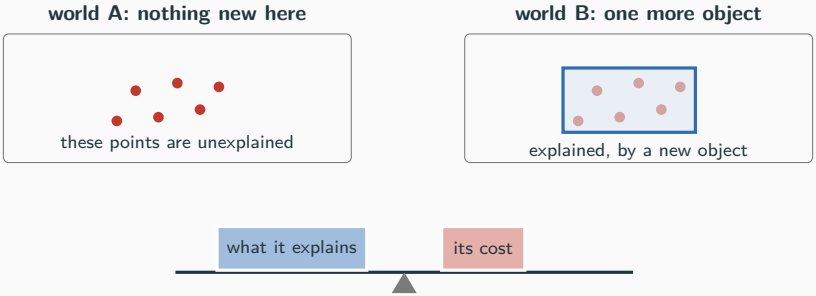
The life of an instance



Pending exists so that a wrong hypothesis costs nothing: nobody has read it yet.
Frozen suppresses an instance's *intake*, not the instance.

source: agents_life_cycle.tex l.379–423

Birth is a comparison between two worlds

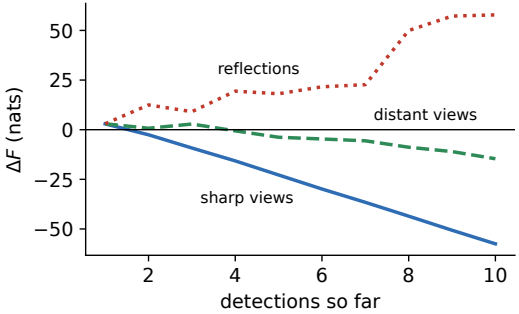


The object is born exactly when what it explains outweighs what it costs, both in nats: the room's corners faced the same trade.

The cost is not a penalty added on top: every live object already pays it.

source: agents_life_cycle.tex l.594–622

A streak cannot read evidence



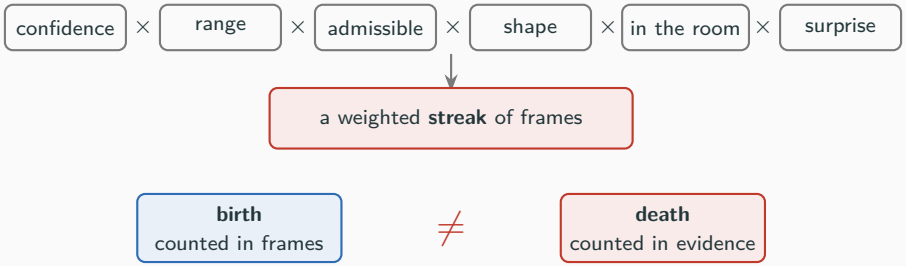
300 runs	streak of 3	$\Delta F < 0$
2 sharp views	0%	98%
10 reflections	94%	0%

A streak counts frames. ΔF reads what each frame says: how sharp it was, and how well the frames agree.

Two sharp views are worth more than ten reflections, and only nats can say so.

source: lab 3, exercises 1–2

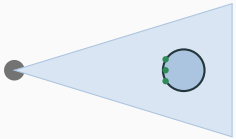
What the robot actually computes: a threshold that should not be one



Each factor is sensible, but the units are gone: birth can be traded against nothing else, so a whole subsystem exists to undo births that should not have happened. The proper comparison is within reach; what is missing is plumbing, not theory.

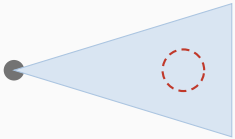
source: agents_life_cycle.tex l.629–706, l.1276–1282

A miss is evidence only where the object would have been seen



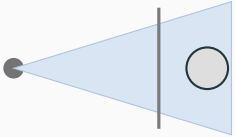
occupancy: a return

confirms



predicted visible, absent

removes



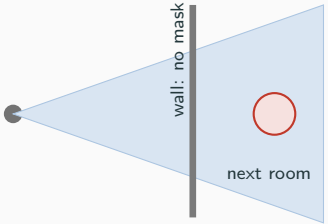
occluded or out of view

holds

The whole stage lies between the middle panel and the right one. Existence is one running number, evidence for and against simply adds to it, and it never reaches certainty, so it can always recover.

source: agents_life_cycle.tex l.887–919

A factor whose argument list was wrong



projects into view, lands on unmasked pixels, finds no mask, votes for its own removal

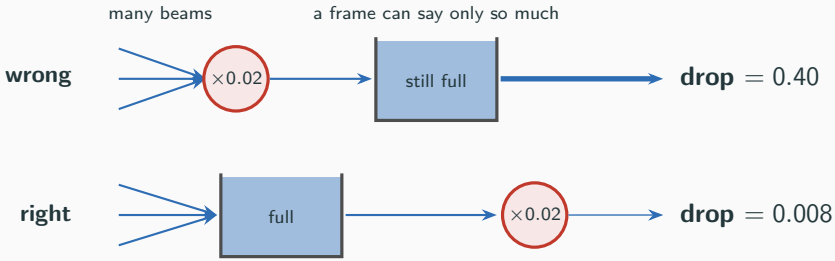
A detector has no class called “wall”, so without the room’s outline every wall is transparent. Line of sight must include the room.

Measured: the same defect, three agents

Refrigerator, then door, then table and cabinet. Each copy was consistent with itself, so no per-agent test could find it.

source: agents_life_cycle.tex l.921–951; CONCEPT_AGENT_LIFECYCLE.md l.7–11

Weight absence by how visible it was, after the bucket

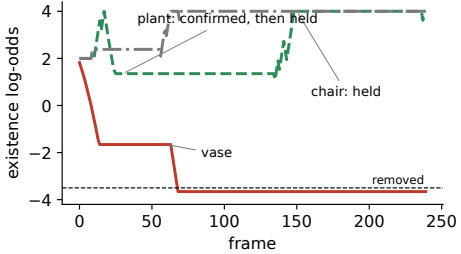


A view in which a present object would be only 2% resolvable.

Turned down before the bucket, the tap does nothing: the bucket fills anyway.

source: agents_life_cycle.tex l.953–973; numbers: lab 3, exercise 4

Four rules, one tour

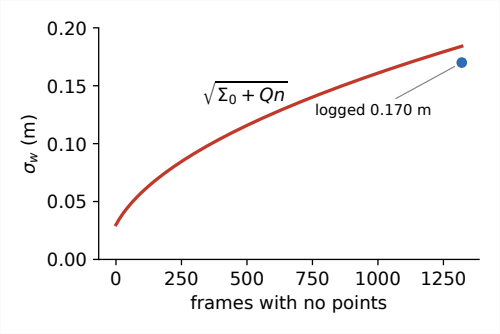


rule	removes (frame)
evidence	vase (68)
no walls in line of sight	vase (68), chair (174)
weight the count	vase (13)
miss counter	everything by frame 44

Never resolvable? No decay constant. The instance is routed to a **verification pull**: uncertainty that cannot be resolved from here is a reason to move.

source: lab 3, exercise 4; agents_life_cycle.tex l.975–985

A constant Q forgets on the agent's clock



Every frame, seen or not, a fixed amount of doubt is added.

- A table known to 3 cm loses half its precision in **3.6 s**.
- It is back at the prior after about **5.9 minutes**, whether it was tracked for thirty seconds or for three hours.

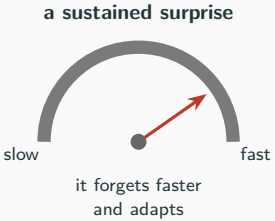
Measured: a real table, 1320 blind cycles

Predicted error 0.184 m; logged 0.170 m. The next sparse mask re-cut the table.

source: agents_life_cycle.tex l.1007–1043

The forgetting rate becomes something the object infers



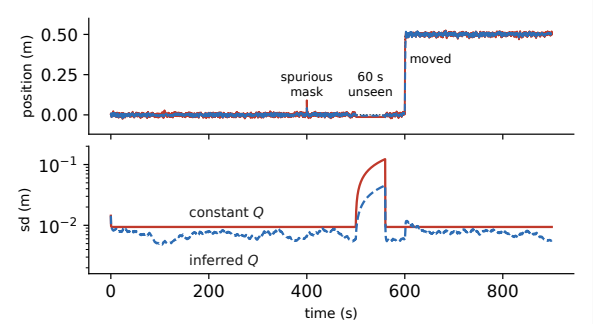


Each object turns its own dial, by how surprised it has been against how surprised it expected to be. One outrageous frame moves it only a bounded amount, and a gentle pull keeps it from sticking: **stiff, but never locked**.

Design only: no agent runs this yet.

source: agents_life_cycle.tex l.1085–1125 (form of the update); lab 3, exercise 5

In the toy: stiff, but not locked



	constant	inferred
sd before (mm)	9.4	6.5
spurious mask (mm)	92	44
after 60 s unseen (mm)	123	45
frames to adapt	9	15

The failure mode: a steady **bias** looks like agreement. A 5 cm bias ends 5.4σ out with constant Q, 7.6σ with inferred.

source: lab 3, exercises 5–6; agents_life_cycle.tex l.1127–1133

Aspect

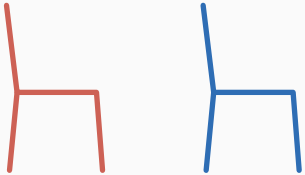
Aspect: the class can be imagined, the instance can be recognised

“How do you imagine a chair?”



the class: what any chair might look like
held inside chair_concept

“Is this the one I saw?”



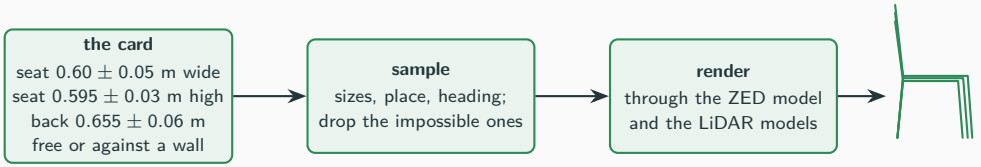
chair_22 chair_23

the instances: what these two looked like, kept with each

The same split as geometry: the class is a prior inside the agent, an instance is a posterior of its own.

Aspect is a dimension of the concept, and it exists as properties of its instances.

How do you imagine a chair? Sample the prior and render it



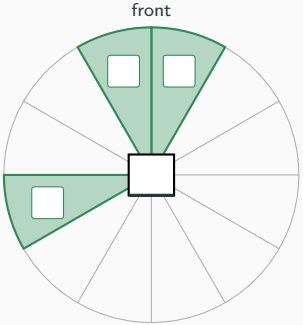
drawn on the chat canvas

- It asks the question "could I see one from there?" of something that is not in the room: the same samples give a probability of detection.
- A chair on the graph is fitted as a pose of a fixed template. Imagination also draws how real chairs vary.
- Today the class's look is a shape, written by hand. No colour, and nothing learned from the chairs it has met.

source: language_agent/src/imagine_view.h; common/concept_manifest/chair.concept.toml

An instance's look is measured one direction at a time

work in progress



12 directions of 30°; the best view of each is kept

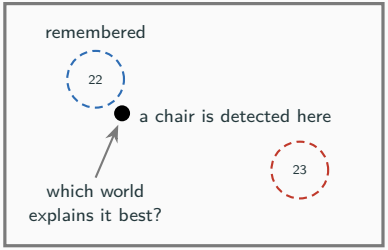
what is kept	where it stands
colour of each instance	runs, for display only; deliberately in no fit
a small crop per direction	on a branch; stored by long-term memory across lives
a learned descriptor per direction (the atlas)	runs in the chair agent in shadow mode; not in long-term memory

A view from the front says little about the back, so a look is a set of views, each tagged with where it was seen from.

Still unreliable: near-identical crops have been filed in far-apart directions.

source: common/appearance_belief; common/view_atlas; branch feat/ltom-thumbnails, 2026-11-09

A second sight is a comparison, and “new” is on the list



is it chair 22?	$\ln p(x \mid 22) + r_{22}$
is it chair 23?	$\ln p(x \mid 23) + r_{23}$
is it **new**?	$\ln(1/A)$

Where it is: the chair stayed, or it was moved with probability π and could be anywhere:

$$p(x \mid k) = (1 - \pi)\,\mathcal{N}(x; \mu_k, \Sigma) + \pi/A.$$

What it looks like: r_k, the nats by which this chair’s views explain the view better than the class does.

Long-term memory only offers. The chair agent decides, the winner must beat “new”, and a tie goes to “new”.

On position alone a moved chair cannot win: $\pi/A < 1/A$.

source: common/recall_match/recall_match.h; common/view_atlas/view_atlas.h

Second sight: what the toy says, what the robot does

work in progress

Toy, % identified correctly

	position	+ look	+ look, same model
chair stayed	98	100	99
chair moved	6	100	33
new chair	81	100	0

With views twice as noisy as the model assumes, the look keeps only 42% of the chairs that never moved. Position alone kept 98%.

Measured: chairs, in simulated tours

Position only, no appearance term. Back in the room, chairs return under their old identity (p .92–.98), also from a previous start of the system.

- The chair is absent for about 35 s after the return, until it is detected again.
- A moved chair is not established: replaying the atlas offline, 15 of 53 right.
- Other classes: merged, never run. Renaming and learned permanence: designed only.

source: lab 3, exercises 18–19; runs ltom_chair2, rose_chair3, 2026-11-08/09

Lab

Lab 3: a concept's life, its footing, its look

`labs/day3.ipynb`, about two hours.

Life cycle

A. **Birth**: ΔF against a streak of three.

B. **Absence**: line of sight through walls; four rules on one tour.

C. **Forgetting**: constant Q against inferred volatility.

Geometry

D. **The room**: the vote, the band, four walls.

E. **The pose**: drive the route; NEES on three maps.

F. **Symmetry**: F at the true pose and its half-turn.

G. **Ceiling**: independent against Woodbury.

Aspect

I. **Second sight**: three remembered chairs and "new"; position against look; one model of chair; a view from the other side.

Short of time: A, B, E, G and I.

Tomorrow

Today we asked what a chair is, how long it lasts and how it looks. Tomorrow: what it is **for**.

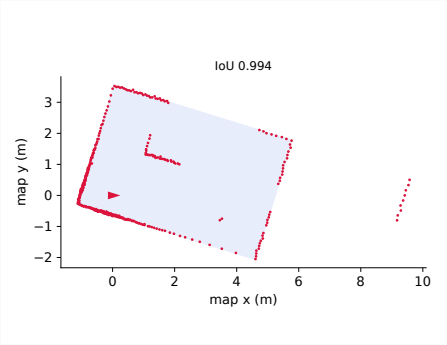
- function: objects publish what they can do for the robot, priced in nats;
- expected free energy: reaching a goal and reducing uncertainty in one currency, and how the two mix;
- the controller: routes and tracking as control problems with no separate obstacle term;
- and the same objective turned on the body: calibration as inference.

Reading for Day 4

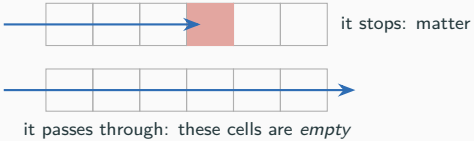
Thesis chapters:

- *Action: affordances and the expected free energy*, up to "Why the gradient decomposes";
- *Self-Calibration of the Robot's Body*: "What the body is", "What no data can determine", "What each motion makes observable".

Backup: in the toy, free space tells a wall from a table



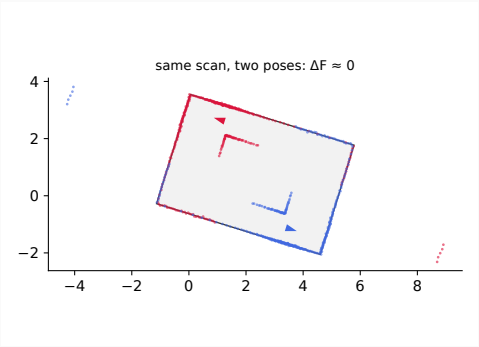
The table's south face earns **82 nats** as a candidate wall, against a price of 7.6. Evidence for a surface cannot keep furniture out of the room.



Only a beam that passes can say a cell is empty. Free floor beyond an edge is evidence against that edge.

source: lab 3, exercise 11; layout_acquisition.tex l.507–526

Backup: Symmetry: where descent cannot help



A rectangle looks the same after a half-turn: the same scan fits both poses with *identical* free energy, and no downhill path joins them.

They are separate hypotheses, so the agent enumerates them and chooses: a discrete search, then local refinement.

This is one of the few places where a threshold is right: a threshold is defensible exactly when the alternative cannot be expressed as a covariance.

source: room_concept.tex l.1869–1927

Backup: Birth is a comparison between two worlds

$$\Delta F = F^{(n+1)} - F^{(n)} = \underbrace{\mathbb{E}_q[-\ln p(\bar{Z} \mid \theta_{1:n+1})] - \mathbb{E}_q[-\ln p(\bar{Z} \mid \theta_{1:n})]}_{\text{accuracy gained by explaining those points}} + \underbrace{D_{\mathrm{KL}}[q(\theta_{n+1}) \,\|\, p(\theta_{n+1} \mid \ell)]}_{\text{complexity paid for the new object}}$$



born exactly when the left side outweighs the right: $\Delta F < 0$

Two models of *different order*, one currency. The complexity is not a penalty added for regularisation: it is the same term every live instance already pays.

source: agents_life_cycle.tex l.594–622

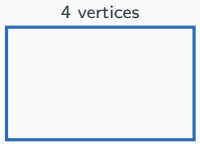
Backup: The room is the first belief born

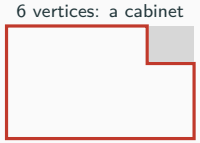
$$F(m) = \underbrace{\mathbb{E}_q[-\ln p(o \mid X, \theta_m, m)]}_{\text{inaccuracy}} + \underbrace{\kappa\,|V|}_{\text{complexity}}, \quad \kappa = \ln \frac{L}{\sigma}$$

The number of walls is unknown, so the room is a choice of *structure* m, priced like a birth. One coordinate to $\sigma = 3$ cm in a room 10 m across costs

$$\ln(10/0.03) \approx 5.8 \text{ nats.}$$

Minimum description length and minimum free energy are, here, one rule.





pays if its cells are worth more than its corners

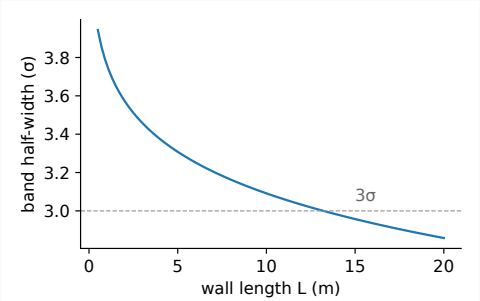
source: layout_acquisition.tex l.56–66, l.128–146

Backup: A band that is derived, not chosen

Clutter is uniform over A. A wall puts its points in a strip L long, σ wide:

$$\frac{p_{\text{wall}}}{p_{\text{clutter}}} = \frac{A}{L\sqrt{2\pi}\,\sigma}\, e^{-r^2/2\sigma^2}$$

- Inlier iff the ratio exceeds 1: about 3σ.
- Each point brings ln(ratio) **nats**.
- The band **narrows as L grows**: a long wall is a more expensive explanation.



A tuned threshold gives the band. The derivation gives the band *and* the currency.

source: lab 3, exercise 8; the same mixture, per cell: layout_acquisition.tex l.1049–1055

Backup: Promotion: the outline with the shortest description



schematic

The outline moves between wall and furniture as the robot drives, so "the latest one" would make the room depend on when exploration stopped. Up to 60 recent outlines are kept and scored on *all* the evidence.

- In 7 of 10 apartment runs (earlier build) the winner was not the last drawn.
- From the kitchen-corner start: overlap 0.968 at promotion, 0.983 refined.
- Ten generated rooms: median 0.994.

All in simulation; the pipeline has not run on the physical robot.

source: layout_acquisition.tex l.1036–1046, l.1173–1177, l.1455–1462

Backup: Observed, precision stops at a floor

One predict plus one update, $\Sigma^{-1} \leftarrow \Sigma^{-1} + i$, with y the predicted precision and $\varrho = Q$:

$$\frac{1}{y} = \frac{1}{y+i} + \varrho \quad \Longrightarrow \quad \boxed{\sigma_{\text{steady}} \approx (\varrho/i)^{1/4}}$$

- Elapsed time does not appear. Evidence is discarded at the rate it is acquired: three hours of observation leave the belief where thirty seconds did.
- Lab: formula 10.0 mm, simulated 9.4 mm.
- Clamping Σ caps the first problem and does nothing about this one.

Nothing an agent learns extends how long it stays resistant.

source: agents_life_cycle.tex l.1045–1082; lab 3, exercise 5

Backup: A constant becomes an inferred quantity

Each instance carries a log-volatility, $Q = e^{\zeta}$, driven by its own surprise:

$$\zeta \leftarrow \zeta + \eta\Big(\underbrace{\tfrac{1}{2}\,\tilde{d} - \tfrac{1}{2}\mathbb{E}[\tilde{d}\,]}_{\text{surprise seen minus expected}} - \underbrace{\frac{\zeta - \zeta_0}{\sigma_\zeta^2}}_{\text{hyperprior pull-back}} \Big)$$

- Consistent views: ζ falls, memory lengthens, precision accumulates.
- A sustained surprise: ζ rises and the model adapts. Stiffness is never a lock.

In the lab $\tilde{d}$ is the saturating (Student-t) surprise of the innovation: one outrageous frame moves ζ by a bounded amount.

Design only: no agent runs this yet.

source: agents_life_cycle.tex l.1085–1125 (form of the update); lab 3, exercise 5

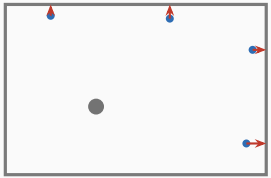
Backup: The room supplies the likelihood

For a scan $\{p_i\}$ and a candidate pose X:

$$d_i(X) = \text{dist}\left(R(\phi)p_i + t,\ \mathcal{R}\right)$$

$$\mathcal{L}_{\text{obs}} = \frac{1}{2\sigma^2} \cdot \frac{1}{N} \sum_i w_i \,\text{Huber}(d_i)$$

- The residual *is* the prediction error: no expected scan is rendered.
- Note the $\frac{1}{N}$: the precision does not grow with scan density.
- Range and incidence weights are precisions on a covariate, not gates.



each return is pulled onto its nearest wall

source: room_concept.tex l.243–307

Backup: What the residual floor is made of



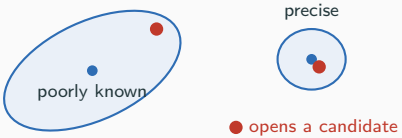
Park the robot. The mean wall distance of each scan hovers near 87 mm and never reaches zero.

- Its fluctuation is white: $N^{-1/2}$, measured ratio 1.41 against a predicted $\sqrt{405/200} = 1.42$.
- That fixes the per-point scatter at ≈ 37 mm, which explains $0.798 \times 37 \approx 30$ mm.

Two-thirds of the floor is an error every return shares. More returns cannot remove it.

source: room_concept.tex l.386–426

Backup: Association: the factor the free energy cannot see



a poorly known instance accepts detections from further away

$$P(M \mid X, Z, \Phi) \approx \mathbf{1}[M = \hat{M}], \quad S = \Sigma^{xy}_{\theta_i} + \sigma^2_{\text{det}} I$$

Squared Mahalanobis distance at most 9, assigned greedily, lowest cost first.

A distribution on one value has no entropy and no expectation: the matching is free, in the literal sense that F cannot see it. A wrong match is invisible to the quantity that would have registered it.

The repair comes later and from elsewhere: merging, and the existence belief.

source: concept_agents.tex l.483–486, l.859–889

Backup: The common-mode ceiling

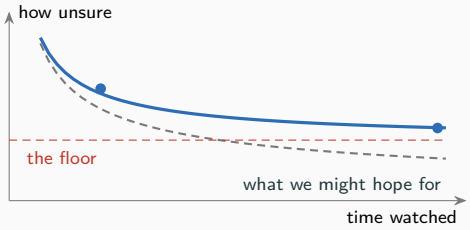
N residuals that share one offset $\delta \sim \mathcal{N}(0, \Sigma_\delta)$, a pose error, plus independent noise. Marginalise δ:

$$\mathcal{I} = J_z^\top \left(\sigma^2 I + J_z \Sigma_\delta J_z^\top\right)^{-1} J_z \overset{\text{Woodbury}}{=} \left(\sigma^2 (J_z^\top J_z)^{-1} + \Sigma_\delta\right)^{-1} \xrightarrow[N\to\infty]{} \Sigma_\delta^{-1}$$

- A denser mask eventually stops helping. The ceiling is set by the *pose* error, not by the pixel count.
- The joint information on (object, offset) has rank 6, not 12: the data cannot tell a displaced object from a displaced robot.

source: concept_agents.tex l.1048–1090

Backup: Precision stops at a floor, in a picture



The doubt added each frame is balanced by what that frame teaches, and the belief stops improving.

Elapsed time does not appear. Three hours of observation leave the belief where thirty seconds did.

Lab: 10.0 mm predicted, 9.4 mm simulated.

Nothing an agent learns extends how long it stays resistant.

source: agents_life_cycle.tex l.1045–1082; lab 3, exercise 5

Backup: The residual floor, in a picture



Park the robot. The average distance from its scan to the walls sits near 87 mm and never reaches zero.

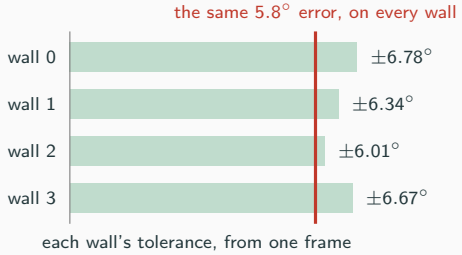
About 30 mm is sensor noise. The other 57 mm is where the map does not match the world, and it is the same on every scan.

Two-thirds of the floor is an error every return shares. More returns cannot remove it.

source: room_concept.tex l.386–426

Backup: Case study: association dies by a common-mode latch

work in progress



This is one **heading** error, not four wall errors, yet each wall is tested on its own against it.

When it crosses about 6°, they all fail in the same frame. Nothing is left to correct the pose, and association goes from 100% to 0% for ever.

"A parked run looks perfect and the first tour kills it."

source: an earlier estimator; post-thesis note, 2026-09-18